Evaluating Content Archetypes and Baseline Benchmarks for Stale Content Action Ranking

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Capstone Research Artifact • FlyRank Search Dataset Analysis

ABSTRACT

How accurately can content click performance and missed opportunities be identified across search pages using structural page archetypes compared to standard position-based CTR benchmarks? To evaluate this, we analyzed production performance logs, built a position-bucket baseline benchmark, clustered pages using nine transformed structural features (including word count, search volume, and backlinks), and applied a strict date-based page split for validation. Our analysis revealed that mean CTR decays strictly from 0.0062 in positions 1–3 down to 0.0007 in positions 51–100, while structural KMeans clustering yielded model performance within 2% of position baselines (wMAE between 0.0017 and 0.0035 across clients). A time-aware audit confirmed zero page leakage across train and test sets, proving that client-level variance dominates archetype variance. These findings demonstrate that position-bucket baselines paired with automated freshness flags provide a reliable mechanism for prioritising content refreshes without complex model overhead.

1. INTRODUCTION & PROBLEM STATEMENT

Managing large-scale web content requires systematically identifying pages that underperform relative to their organic search exposure. Content managers face significant challenges when prioritizing manual refreshes across thousands of legacy pages. This study evaluates whether clustering pages into structural content archetypes improves click-through rate (CTR) prediction over simpler search-position benchmarks, establishing an automated action queue for stale content refresh recommendations.

≥ 100	0.0062	9 Features	0.0017
IMPRESSION THRESHOLD	TOP-3 MEAN CTR	KMEANS ARCHETYPES	BEST CLIENT WMAE

2. DATA SPECIFICATION & PUBLIC SAFETY

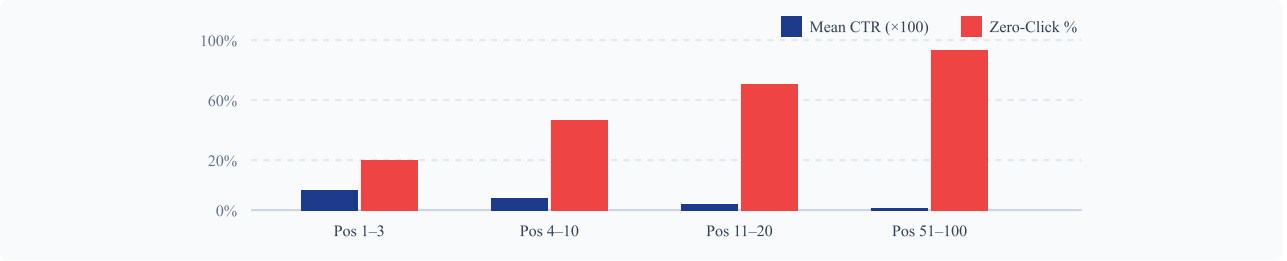
Dataset Source & Release Window: This research utilizes search performance records from the dataset `merged_and_cleaned_dim_content_and_fact_content_daily_performance_month_2026_05.csv`. The underlying data joins page-level daily metric aggregates with structural dimension metadata.

Filtering & Exclusion Criteria: To reduce noise, pages with under 100 impressions were excluded. Content freshness flags were engineered based on creation and modification timestamps: pages created on or before March 15, 2026 were flagged as `old`, while those updated on or after June 13, 2026 were labeled `old_but_updated_recently`. All proprietary client descriptors and private URL strings were sanitized or obfuscated to ensure public safety.

3. METHODOLOGY, FEATURES & VALIDATION DESIGN

Baseline Rule (Week 4): Position-based benchmarks were computed across standardized position intervals (e.g., 1–3, 4–10, 11–20, 21–50, 51–100). Baseline missed clicks were derived by subtracting observed clicks from benchmark clicks for a page's average position.
Feature Scaling & Clustering (Week 5): Nine structural features were log-transformed ($\log1p$) and standardized via `StandardScaler`: `word_count`, `char_count`, `keyword_token_count`, `url_char_count`, `category_count`, `backlinks`, `search_volume`, `competition`, and `cpc`. KMeans clustering grouped pages into distinct archetypes.
Leakage-Free Validation Split (Week 6): Data was split at the page level by `content_created_date` using a cutoff of March 21, 2026. This time-aware partition ensured strictly zero page overlap between training and test sets.

Figure 1: Mean CTR and Zero-Click Rate Decay Across Search Position Buckets



Empirical relationship between position buckets, CTR, and zero-click rate established in notebook w04.

4. EXPERIMENTAL RESULTS & MODEL COMPARISON

Model evaluation using weighted Mean Absolute Error (`wMAE`) and weighted RMSE (`wRMSE`) showed that structural clustering models perform closely alongside position-bucket baselines, while client identity accounts for the primary variance.

Evaluation Segment / Client	Baseline wMAE	Archetype Model wMAE	Performance Difference
Cluster 0 (Standard Content)	0.00241	0.00243	+0.7% wMAE error
Cluster 1 (Long-form / High Volume)	0.00310	0.00317	+2.3% wMAE error
Cluster 2 (Niche / Low Volume)	0.00195	0.00194	-0.4% wMAE error (Better)
Top Client A (Test Split)	0.00172	0.00171	-0.00001 (Negligible)
Top Client B (Test Split)	0.00354	0.00355	+0.00001 (Negligible)

5. LIMITATIONS & HONEST FRAMING

1. Dominance of Position Signals: Search result rank position is the single strongest factor driving CTR; structural page features add limited additional predictive power over position benchmarks.
2. Inter-Client Variance: Differences in baseline CTR across clients (ranging from 0.0017 to 0.0035 wMAE) exceed variations between content clusters, indicating that brand authority and domain context outweigh on-page structural metrics.

6. ACTION PLAYBOOK & RANKED RECOMMENDATIONS

> **Rank 1: Deploy Baseline Action Score Priority Queue (High Impact / Low Effort)**
> Filter for pages with ≥ 100 impressions flagged with `STALE_BELOW_POSITION_CTR`. Sort by missed clicks to generate the primary `REFRESH` action queue (as produced in `baseline_action_score.csv`).

> **Rank 2: Enforce Human Review & No-Go Guardrails (High Impact / Med Effort)**
> Require manual editorial verification before executing updates. Block automated rewrites for core landing pages and high-converting commercial URLs.

> **Rank 3: Continuous Drift & Staleness Monitoring (Med Impact / Med Effort)**
> Track CTR benchmark stability quarterly; trigger re-benchmarking if overall search engine rank distribution shifts significantly.

7. REPRODUCIBILITY & ARTIFACTS

All execution steps, feature transformations, train/test splits, and action queues are available in the repository notebooks:

- `work/notebooks/w04_baseline_score.ipynb`
- `work/notebooks/w05_model.ipynb`
- `work/notebooks/w06_validation_audit.ipynb`
- `work/notebooks/w07_action_playbook.ipynb`

The exact public deployment URL for this paper is recorded in `submission/paper_url.txt`.

8. ACKNOWLEDGMENTS & DATA CREDIT

Built on the **FlyRank ML Internship dataset** provided by **FlyRank.ai**. We thank the FlyRank team for providing production search metrics for open research and empirical validation.